# Teacher Guide — primary / module-09-subtraction-nikhilam

IKS Curriculum

## Table of Contents

## Overview

## Module 9 — Friends to Ten! (Towards *Nikhilam*)

**Age band:** Primary · **Grades:** 3–5 · **Duration:** 10 sessions × 35 min

### What children will learn

1. The "friends to ten" idea — every single-digit number has a friend that completes it to 10. (3's friend is 7. 8's friend is 2.)
2. To compute `100 − a single-digit number` instantly. (`100 − 7 = 93`.)
3. The Sanskrit name of the rule: *Nikhilam* (introduced gently as "the friends rule").
4. To find the "friend to 100" of a two-digit number with help.
5. A peek at the dot-method's sister technique through bar models and stories.

### How this module sounds

- Lots of stories. Friends, hugs, completing pairs.
- Bar models drawn in colour — long bars of 10 or 100.
- Sanskrit term *Nikhilam* introduced on Day 1 as the magic word; not memorised in full.
- No long worksheets. Children move, talk, draw, count out loud.

### NEP 2020 alignment

- §5.1 — Foundational numeracy.
- §4.6 — Indian Knowledge Systems.
- §4.23 — Experiential learning.

### Materials

- Number cards 0–9 (4 sets per pair) — same as Module 5
- Bar-model paper (long strips of paper, or printed bars)
- Coloured pencils
- Pretend currency: paper 10s and 100s
- "Friends to Ten" wall poster

## Day-by-day

| Day | Title | What we do |
|---|---|---|
| 1 | Friends to Ten | Story + chant + card game |
| 2 | Hugs to Ten | Pair drill: who completes whom? |
| 3 | The Magic Word | Introduce *Nikhilam* in story |
| 4 | 100 minus Friends | `100 − 7 = 93`, the bar-model way |
| 5 | The Coin Game | Shopkeeper play with paper money |
| 6 | All from 9, Last from 10 | Gentle rule introduction |
| 7 | Big-Friends Day | Find friends-to-100 of bigger numbers |
| 8 | Project Day 1 | Build your own "Friends to Ten" game |
| 9 | Project Day 2 | Finish + test |
| 10 | Showcase | Play each other's games |

## Project

Children design their own "Friends to Ten" game in pairs. Cards, board, dice — anything. They play classmates' games on Day 10.

See `assessments/project-brief.md`.

## Sources

The "friends to ten" pattern is foundational arithmetic that every culture's basic teaching includes. The Sanskrit name *Nikhilam* and its formal sūtra form come from Tirthaji's *Vedic Mathematics* (1965). At this band we plant the seed; the full method comes in Middle band. For teacher reference, see `sources.md`.

# 10-Day Lesson Plan

# Module 9 (Primary) — 10-Day Lesson Plan

## At a glance (35-min sessions)

| Day | Title | Activity | Output |
|---|---|---|---|
| 1 | Friends to Ten | Story + chant + Make-Ten | First-day fun |

| Day | Title | Activity | Output |
|---|---|---|---|
| | | card game | |
| 2 | Hugs to Ten | Pair drill | Each child names all 5 friends |
| 3 | The Magic Word | Introduce *Nikhilam* via story | Chant the word 5× |
| 4 | 100 minus Friends | `100 − 7 = 93`, bar-model | Drawing |
| 5 | The Coin Game | Shopkeeper play | Game card |
| 6 | All from 9, Last from 10 | Gentle rule, with hand actions | Mini-poster |
| 7 | Big-Friends Day | Friends-to-100 of 2-digit numbers | Worksheet |
| 8 | Project Day 1 | Game design | Sketch + first prototype |
| 9 | Project Day 2 | Build + test | Game ready to play |
| 10 | Game Showcase | Play each other's games | Stickers |

## Daily structure (35 min)

| Time | What |
|---|---|
| 5 min | Settle + the "Friends to Ten" chant |
| 8 min | Story / new game introduction |
| 17 min | Game / activity |
| 5 min | Sharing / reset |

## The daily chant

> "1 and 9 are friends! 2 and 8 are friends! 3 and 7 are friends! 4 and 6 are friends! 5 and 5 are friends! All make TEN!"

Sing it / chorus it with claps. By Day 3 it should be effortless. Same chant as Module 5; children come in with it half-learned.

## Pacing notes

- Games > worksheets at this band. If a worksheet feels slow, drop it and add another game round.
- Day 3 (the *Nikhilam* magic word) is a "wow" moment — keep it short and mysterious.
- Day 6 introduces "all from 9, last from 10" — but only verbally, with hand actions. Don't push for full mastery; that's a Middle-band goal.
- Days 8–10 build to the game showcase. Confirm groups by Day 6.

## What children leave with

- Instant recall of all 5 friends-to-ten.
- A mental habit of "completing" when subtracting.
- The Sanskrit word *Nikhilam* — pronounced and recognised.

- One game they made and one they played.

# Teacher Notes

# Teacher Notes — Module 9 (Primary)

## Posture

Friends-to-ten fluency is the single skill that pays off everywhere later. This is what we're building. The Sanskrit name *Nikhilam* and the rule "all from 9, last from 10" are *seeds* — children meet them, can pronounce them, recognise them when they re-appear in Middle band. We do not push for mastery of the full rule at this band.

## Common things to handle

1. **A child can't yet recall all 5 friends-to-ten.** That's fine — the module is the practice. Don't shame; bring them into more games.
2. **A child gets bored ("I already know my friends!").** Give them the "subtraction" challenge: `100 - 7 = ?` Or `100 - 23 = ?` with bar models.
3. **A child says "But what about negative numbers?"** Acknowledge — *"That's a question for older grades. Today we play with the counting numbers."*
4. **A child mispronounces *Nikhilam*.** That's fine. The goal is exposure, not perfection. Model it slowly: ni-khi-lam. Three soft syllables.
5. **A child asks "Is this REALLY from the Vedas?"** Honest answer: *"A wise teacher named Tirthaji wrote a book in 1965 that gave this rule. Some people think it's older. We don't know for sure. The rule works — that's what we care about today."*

## Safety

- No physical risk in this module.
- Pretend currency activities: keep it light; some children's families have money stress. Frame as game-money.

## Differentiation

- **Tier up:** Grade 5 children may be ready for 2-digit + 2-digit complements to 100. Push to `100 - 47 = 53` by Day 7. Show them the Middle-band rule briefly.
- **Tier down:** Grade 3 children building friends-to-ten fluency need EXTRA game time. Skip the bar-model drawing if it slows them; do another card game instead.

## Project oversight

- Days 8–10: children design a "Friends to Ten" game. Encourage variety — card games, board games, dice games, jumping games.
- Materials limit: only what's in the classroom. Don't expect families to provide game supplies.

- Theme must be visible — somewhere in the rules, children must use friends-to-ten or completing to 100.

### Honest framing

Some parents will arrive convinced *Nikhilam* is “ancient sacred wisdom.” Don’t combat this aggressively at this band. Just be clear: the rule is in a 1965 book by Tirthaji; we teach the rule; we save the historicity question for older grades. Use the parent-guide language.

### If you only have 6 days

- Day 1: Friends to Ten game + chant
- Day 2: 100 minus friends + The Magic Word (combined)
- Day 3: Coin Game (Day 5 material)
- Day 4: Project sketch
- Day 5: Project build
- Day 6: Showcase

Skip the bar-model drawing day.

# Sources

# Sources — Module 9 (Primary)

At this band, sources are teacher reference only. Children don’t cite anything in their work.

### Source for the rule

- Tirthaji, *Vedic Mathematics* (1965), sūtra #2: *Nikhilaṁ Navataścaramaṁ Daśataḥ* — “all from nine and the last from ten.”
- The Sanskrit phrase is in the SutraGanita corpus PDFs: *Vedic Mathematics.pdf*, *Microsoft Word - sutras1.pdf*, and others in `/Users/sree/Projects/SutraGanita/content/`.

### Why we don’t teach the full rule at Primary

The full Nikhilam method requires:

- Fluent friends-to-ten (foundational — this is what we drill at Primary).
- Place-value understanding (some Grade 3, most Grade 5).
- Comfort with subtraction across two-digit numbers.

Most Grade 3–4 children are still building the first. We focus there and tease the rest. The Middle-band version teaches the full method to Grade 6–8.

## Honest framing for families who ask

The Nikhilam method is from a 20th-century book — *Vedic Mathematics* by Bharati Krishna Tirthaji (1965). Tirthaji presented his 16 sūtras as drawn from the Vedas; no other Sanskritist has confirmed that source. We teach the method enthusiastically and are honest about its history: the technique is real, the deeper Vedic attribution is unverified. By Grade 9 students read the scholarly debate.

## For teacher reference

- *FUNDAMENTAL AND VEDIC MATHEMATICS .pdf* — chapter on Nikhilam with worked examples (teacher background).
- The Middle-band module (`curriculum/middle/module-09-subtraction-nikhilam/`) is a useful reference for what comes next.

# Day 1 — Friends to Ten

**Module:** Friends to Ten (Primary) · **Time:** 35 min

## Goal

Children meet "friends to ten" through a chant, a story, and a card game. By end of class they can name at least three friend-pairs.

## Materials

- A deck of number cards 0–9 (4 of each = 40 cards) per pair
- Drawing paper + crayons
- "Friends to Ten" wall poster (5 colourful pairs)

## The flow

### Settle + chant (5 min)

Sit on the mat. Teacher: > *"For the next two weeks we'll meet some friends. They're tiny — but very important. Numbers have friends too."*

Teach the chant (3 times, claps on each pair):

> "1 and 9 are friends! 2 and 8 are friends! 3 and 7 are friends! 4 and 6 are friends! 5 and 5 are best friends! All make TEN!"

### Story (5 min)

> **The Friend Game**
>
> Ten little ants lived on a leaf. Sometimes they walked off in groups.
>
> One day, 3 ants walked off. *(Hold up 3 fingers.)* How many were left behind on the leaf? *(Wait. Children call out: 7!)*

Yes! 3 and 7 are friends. They always make 10 together.

Another day, 6 ants walked off. *(Hold up 6.)* How many on the leaf? *(4!)*

6 and 4 are friends. Always 10.

Every number from 0 to 10 has a friend. We'll meet them all today.

### Card game: Make Ten (20 min)

**How to play (paired):**

1. Shuffle. Deal 5 cards to each player. Put 5 face-up in the middle.
2. On your turn:
    - Look at your hand and the middle.
    - Find two cards that are friends (add to 10).
    - Take both. Put in your pile.
    - Draw 1 from the deck.
3. If you can't make 10, place 1 card in the middle. End turn.
4. When the deck runs out, count pairs. Most pairs wins.

Play 2–3 rounds.

### Show + chant (5 min)

- "Who won a round?" — winner shares ONE friend pair they made.
- Class chants the pair: *"6 and 4 are friends!"*
- Close with the daily chant.

## Homework / take-home

- Practise the chant with a sibling or parent. Win a round of Make Ten with family!

## Differentiation

- **Tier up:** Use 0–9 cards including 0+10 pairs.
- **Tier down:** Use cards 1–6 to keep pairs simple.

## Teacher's quick notes

- Some children won't find pairs and will give up. Pair them with a stronger player as a "team."
- Same game as Module 5 — children may come in already half-knowing it. Affirm and add the *friend* framing.
- The "friend" language is the bridge to subtraction. By Day 4 we'll say "to find 100 minus 7, ask: who's 7's friend? 3. Then put a 9 in front. 93."

## Teacher note on Sanskrit

We don't introduce *Nikhilam* yet. The word appears on Day 3 as a "magic word." Today is just about friends.

# Day 2 — Hugs to Ten

**Module:** Friends to Ten (Primary) · **Time:** 35 min

## Goal

Children practise friend recall in a paired drill until every child can name a friend in under 2 seconds.

## Materials

- Number cards 0–9 (one set per pair)
- Stopwatch (optional)

## The flow

### Settle + chant (5 min)

The friends-to-ten chant, 3 times.

### Story-warm-up (3 min)

> *"Yesterday we met the friends. Today they want to HUG. When two friends meet, they hug and become TEN."*

Demonstrate: hold up a "3" card. Pause. *"Who's 3's friend?"* Children chorus: SEVEN! Bring out a "7" card. Hug them together. *"Now they're TEN."*

### Paired drill (20 min)

**Round 1 — Quick recall (8 min):** - One child holds the deck. Flips card. - Other child names the friend. - Stack hits / misses in two piles. - Swap roles after 30 seconds. - Goal: 10 right in a row.

**Round 2 — Speed (5 min):** - Time how many friends you can name in 30 seconds. - Both children try.

**Round 3 — Hide the friend (7 min):** - Lay 5 cards face-up. Hide one (turn over). - Partner has to GUESS the hidden card based only on its friend. - E.g., visible cards are 1, 3, 5, 7, 9; sum should be 25. Actual sum visible = ? Difference tells you the hidden card. - (For Grade 3, simplify: just show one card's friend; partner names what's hidden.)

### Wrap-up (7 min)

- Quick chorus: teacher calls a digit; class chorus the friend. Do 15 in 60 seconds.
- 4 → SIX! 8 → TWO! 1 → NINE! 6 → FOUR! 2 → EIGHT! 5 → FIVE! …
- Daily chant.

## Homework / take-home

- Friend drill with family: 20 digits in 60 seconds. Track how many you got.

## Differentiation

- **Tier up:** “Friends to 100” preview: who completes 73? (27.) Show the friend on bar paper.
- **Tier down:** Use only 1–5 cards. Friends will be larger but the count is shorter.

## Teacher’s quick notes

- The HUG metaphor is sticky for this age group. Hold it.
- Some children laugh at “5 and 5 are best friends.” Let them. The laugh helps the math stick.
- Avoid timed competition between pairs publicly. Each pair races itself.

# Day 3 — The Magic Word

**Module:** Friends to Ten (Primary) · **Time:** 35 min

## Goal

Children meet the Sanskrit word *Nikhilam*. They learn to pronounce it, recognise the wall poster, and connect it to “friends to ten.”

## Materials

- A new wall poster: the word ***Nikhilam*** in big letters, with a smiling face. Underneath, in small letters: “the friend rule.”
- Drawing paper
- Crayons

## The flow

### Settle + chant (5 min)

The friends-to-ten chant, 3 times.

### The magic-word story (10 min)

> **The Sanskrit Word**
>
> Long ago, people in India had a special name for “the friend rule.” The name is a Sanskrit word.
>
> *(Pause for suspense.)*
>
> Are you ready?
>
> The word is — **Nikhilam**! *(Say it slowly: ni-khi-lam.)*
>
> *(Class echoes 3 times. Soft / normal / loud. Children find it funny — let them.)*
>
> *Nikhilam* means “all of it” or “the whole.” It’s the secret name for our friends-to-ten rule.

> An older book from 1965 gives this rule a fancy name. We'll learn the long name in Grade 6. For today — just say *Nikhilam*. The friend rule. The whole. All of it.

### Drawing time (15 min)

Each child draws ONE friend-pair. Above the pair, write the word *NIKHILAM* in big letters (or use the wall poster as a guide).

Examples: a pair of trees labelled "3 and 7." Two children holding hands labelled "4 and 6." Two suns labelled "5 and 5."

(Teacher walks around helping with the spelling of "Nikhilam.")

### Sharing (5 min)

3 children hold up their drawings. Class chorus the friend-pair AND chants *Nikhilam* together.

End with the daily chant + one big "*NIKHILAM!*"

## Homework / take-home

- Tell a parent the magic word: *Nikhilam.* Say it 5 times. Win their amazement.

## Differentiation

- **Tier up:** Try to write *Nikhilam* in Devanāgarī (निखिलं). Provide a print-out card.
- **Tier down:** Just say the word. Drawing is optional.

## Teacher's quick notes

- The pronunciation is harder than it looks. Model slowly. Don't correct individual children publicly — they'll get it by Day 6 through repetition.
- Some children will ask "what does it mean?" Answer: "all" or "the whole" or "every part." The full meaning becomes clear in Middle band.
- This is the first Sanskrit word many children will say with confidence. Honour it. Some will go home and say it 200 times.

## Citation(s) used (teacher reference, not student-facing)

- Tirthaji, *Vedic Mathematics* (1965), sūtra Nikhilam.

# Day 4 — 100 minus Friends

**Module:** Friends to Ten (Primary) · **Time:** 35 min

## Goal

Children compute `100 - N` for single-digit N using "friends to ten" and a bar model. By end of class: `100 - 7 = 93`, instantly.

## Materials

- Bar-model strips: long printed strips divided into 10 boxes of 10 each (= 100 squares total)
- Coloured crayons
- Number cards 0–9

## The flow

### Settle + chant (5 min)

The friends-to-ten chant. A new line: “*And TEN tens make ONE HUNDRED!*”

### Bar-model demo (10 min)

Draw a big bar on the board (or use the strip). Mark off 10 sections of 10 each. Total: 100.

Teacher: *“100 ants on a hundred leaves. 7 ants walk off. How many left?”*

Cross off 7 squares from the rightmost group. *“This last group used to be 10. Now it’s 3.”*

*“How many groups of 10 still left? NINE groups. Plus the 3 leftover ants. So 90 + 3 = 93.”*

> **The trick:** 7’s friend is 3 (10 minus 7 = 3). And there are 9 groups of 10 still standing. So `100 − 7 = 93`.

Demo 3 more, with class predicting:

| Problem | Friend of N | 9 in front | Answer |
|---|---|---|---|
| 100 − 4 | 6 | 9 | **96** |
| 100 − 8 | 2 | 9 | **92** |
| 100 − 1 | 9 | 9 | **99** |
| 100 − 9 | 1 | 9 | **91** |

### Activity: bar-model worksheet (15 min)

Each child gets a printed worksheet with 6 bar-model problems:

```
100 − 3 = ?
100 − 5 = ?
100 − 6 = ?
100 − 2 = ?
100 − 8 = ?
100 − 7 = ?
```

For each: shade the right number of squares, then read off the answer. Bonus: write the friend pair next to it.

### Wrap-up (5 min)

- Quick fire: teacher calls `100 − N`; children chorus the answer.

- Daily chant.

### Homework / take-home

- 10 problems: `100 − 4, 100 − 9, 100 − 6, 100 − 2, 100 − 5, 100 − 8, 100 − 3, 100 − 7, 100 − 1, 100 − 0`. Aim for under 30 seconds.

### Differentiation

- **Tier up:** Try `100 − 10`, `100 − 15`, `100 − 20`. Bar model still works.
- **Tier down:** Use only N from 1 to 5. Build up.

### Teacher's quick notes

- The "9 in front" framing is the key insight. Make sure every child says it.
- For N = 0 or N = 10, the bar model is a bit off (no shading, or a whole group gone). Acknowledge and move on; those are edge cases.
- Some children will sniff out the pattern: "to do 100 minus any single digit, just write '9' then the friend of that digit." Praise — that IS the Nikhilam rule for `100 − N`.

# Day 5 — Why It Works — The Algebra

**Module:** Subtraction — Nikhilam (All from 9) (Primary) · **Time:** 30 min

### Goal

Derive the place-value identity that makes the rule work.

### Materials

- Notebook, pencils, drawing paper
- (See lesson-plan.md for any day-specific materials)

### The flow

#### Settle + chant (3 min)

Daily chant or breathing exercise.

#### Story recall (4 min)

Briefly recall yesterday's lesson by asking: *"What's one thing you remember from yesterday?"*

#### Today's idea (10 min)

- *Nikhilaṁ Navataścaramaṁ Daśataḥ* (IAST: Nikhilaṁ Navataścaramaṁ Daśataḥ; lit. ""all from 9, the last from 10"")

- Why does the trick work? Let's see!

Write 100 = 99 + 1. - 99 − 37 is easy (no borrow): 6 from 9 = 6 and 9 - 3 = 6, write it out as: 9-3 = 6, 9-7 = 2… actually let's keep it simple. - 100 − 37 = 63. - And 9-3 = 6, 10-7 = 3 → 63.

The 9s help because there's no borrow. *"That's the magic!"*

### Activity (8 min)

A simplified version of *Complement Flashcards* (see `activities/`). Draw, sing, or act it out.

### Wrap-up (3 min)

Each child shares one word that describes today.

## Homework

(See `homework.md`.)

## Teacher notes

- Citations belong in teacher notes only, not in student-facing material at this band.
- Keep new Sanskrit terms to one per lesson, repeated several times.
- If energy is low, swap the activity for free drawing.

# Day 6 — Bar Model Proof

**Module:** Subtraction — Nikhilam (All from 9) (Primary) · **Time:** 30 min

## Goal

Draw a visual proof using bar models and number lines.

## Materials

- Notebook, pencils, drawing paper
- (See lesson-plan.md for any day-specific materials)

## The flow

### Settle + chant (3 min)

Daily chant or breathing exercise.

### Story recall (4 min)

Briefly recall yesterday's lesson by asking: *"What's one thing you remember from yesterday?"*

### Today's idea (10 min)

- *pratiyogi (modern coinage)* (IAST: pratiyogi (modern coinage); lit. "the number that adds to a base")

- Let’s see the trick with bars (or paper strips)!

Cut a strip 100 cm long. Now cut off 37 cm. *“How long is the rest?”* (63 cm.)

Now cut a strip 99 cm. Cut off 37 cm. *“How long?”* (62 cm — easy! 9-3=6, 9-7=2.) Add 1 cm — 63 cm.

*“Same answer! Easier with 99!”*

### Activity (8 min)

A simplified version of *Complement Flashcards* (see `activities/`). Draw, sing, or act it out.

### Wrap-up (3 min)

Each child shares one word that describes today.

## Homework

(See `homework.md`.)

## Teacher notes

- Citations belong in teacher notes only, not in student-facing material at this band.
- Keep new Sanskrit terms to one per lesson, repeated several times.
- If energy is low, swap the activity for free drawing.

# Day 7 — Two’s Complement in Computers

**Module:** Subtraction — Nikhilam (All from 9) (Primary) · **Time:** 30 min

## Goal

Show how the same idea underlies binary computer subtraction.

## Materials

- Notebook, pencils, drawing paper
- (See lesson-plan.md for any day-specific materials)

## The flow

### Settle + chant (3 min)

Daily chant or breathing exercise.

### Story recall (4 min)

Briefly recall yesterday’s lesson by asking: *“What’s one thing you remember from yesterday?”*

### Today’s idea (10 min)

- *Nikhilaṁ Navataścaramaṁ Daśataḥ* (IAST: Nikhilaṁ Navataścaramaṁ Daśataḥ; lit. ““all from 9, the last from 10”“)

- Computers use the same trick! In computers, numbers are 0s and 1s. They use “all from 1, last from 1+1” — same idea but with 1 instead of 9.

Show a calculator. *“When you press 100 − 37, the computer secretly uses our trick to find the answer fast.”*

Children share: *“What’s the coolest computer thing they know?”*

### Activity (8 min)

A simplified version of *Complement Flashcards* (see `activities/`). Draw, sing, or act it out.

### Wrap-up (3 min)

Each child shares one word that describes today.

## Homework

(See `homework.md`.)

## Teacher notes

- Citations belong in teacher notes only, not in student-facing material at this band.
- Keep new Sanskrit terms to one per lesson, repeated several times.
- If energy is low, swap the activity for free drawing.

# Day 8 — Speed Day

**Module:** Subtraction — Nikhilam (All from 9) (Primary) · **Time:** 30 min

## Goal

Complete a 30-problem mental subtraction sprint.

## Materials

- Notebook, pencils, drawing paper
- (See lesson-plan.md for any day-specific materials)

## The flow

### Settle + chant (3 min)

Daily chant or breathing exercise.

### Story recall (4 min)

Briefly recall yesterday's lesson by asking: *"What's one thing you remember from yesterday?"*

### Today's idea (10 min)

- *pratiyogi (modern coinage)* (IAST: pratiyogi (modern coinage); lit. "the number that adds to a base")
- Sprint day! 10 problems in 3 minutes.

Each child gets a worksheet: 1. 10 − 4 2. 100 − 37 3. 100 − 62 4. 1000 − 250 5. 100 − 88 …

Cheer for everyone who finishes. *"Speed isn't everything — accuracy matters more!"*

### Activity (8 min)

A simplified version of *Complement Flashcards* (see `activities/`). Draw, sing, or act it out.

### Wrap-up (3 min)

Each child shares one word that describes today.

## Homework

(See `homework.md`.)

## Teacher notes

- Citations belong in teacher notes only, not in student-facing material at this band.
- Keep new Sanskrit terms to one per lesson, repeated several times.
- If energy is low, swap the activity for free drawing.

# Day 9 — Mixed Practice

**Module:** Subtraction — Nikhilam (All from 9) (Primary) · **Time:** 30 min

## Goal

Solve a mixed problem set comparing complement vs traditional borrow.

## Materials

- Notebook, pencils, drawing paper
- (See lesson-plan.md for any day-specific materials)

## The flow

### Settle + chant (3 min)

Daily chant or breathing exercise.

### Story recall (4 min)

Briefly recall yesterday's lesson by asking: *"What's one thing you remember from yesterday?"*

### Today's idea (10 min)

- *Nikhilaṁ Navataścaramaṁ Daśataḥ* (IAST: Nikhilaṁ Navataścaramaṁ Daśataḥ; lit. ""all from 9, the last from 10"")

- Today: pick the right tool!

Show 10 problems on cards. For each, children hold up a sign: - "Trick!" (for subtracting from 10, 100, 1000) - "Regular!" (for harder cases)

Then solve 5 trick-type problems together.

### Activity (8 min)

A simplified version of *Complement Flashcards* (see `activities/`). Draw, sing, or act it out.

### Wrap-up (3 min)

Each child shares one word that describes today.

## Homework

(See `homework.md`.)

## Teacher notes

- Citations belong in teacher notes only, not in student-facing material at this band.
- Keep new Sanskrit terms to one per lesson, repeated several times.
- If energy is low, swap the activity for free drawing.

# Day 10 — Final Assessment

**Module:** Subtraction — Nikhilam (All from 9) (Primary) · **Time:** 30 min

## Goal

Take the summative quiz.

## Materials

- Notebook, pencils, drawing paper
- (See lesson-plan.md for any day-specific materials)

## The flow

### Settle + chant (3 min)

Daily chant or breathing exercise.

### Story recall (4 min)

Briefly recall yesterday's lesson by asking: *"What's one thing you remember from yesterday?"*

### Today's idea (10 min)

- *pratiyogi (modern coinage)* (IAST: pratiyogi (modern coinage); lit. "the number that adds to a base")

- Final day! We celebrate the subtraction trick.

Each child shows one favourite trick problem to a partner. Cheer for each.

Take the small quiz with stickers. Every child gets a "Subtraction Sleuth" sticker.

End: chant — *"All from nine, last from ten!"*

### Activity (8 min)

A simplified version of *Complement Flashcards* (see `activities/`). Draw, sing, or act it out.

### Wrap-up (3 min)

Each child shares one word that describes today.

## Homework

(See `homework.md`.)

## Teacher notes

- Citations belong in teacher notes only, not in student-facing material at this band.
- Keep new Sanskrit terms to one per lesson, repeated several times.
- If energy is low, swap the activity for free drawing.

# Formative Quiz — Subtraction — Nikhilam (All from 9) (Primary)

**When:** mid-module (after Day 5). **Duration:** 20 minutes. **Format:** mix of short answer, multiple choice, and reflection. Sticker self-assessment for Primary.

## Questions

1. Name the three / four key terms introduced in this module so far. Give the IAST and one-line meaning of each.

2. State the central idea of Subtraction — Nikhilam (All from 9) in one sentence.
3. (Multiple choice) Which day in this module focused on 'Subtract from a Power of 10'? (a) Day 1 (b) Day 2 (c) Day 3 (d) Day 4
4. True / False — and explain in one sentence: *Subtraction — Nikhilam (All from 9) is purely a religious concept and has no scientific framing.*
5. Connect Subtraction — Nikhilam (All from 9) to **one** other module in this curriculum. Which module, and how?
6. List **two** sources cited so far in this module.
7. (Short answer) Why does the module distinguish 'Nikhilaṁ Navataścaramaṁ Daśataḥ' from related concepts? Give one example.
8. (Short answer) In your own words, explain Day 3's concept.
9. Identify **one** common misconception about Subtraction — Nikhilam (All from 9) and rebut it in one sentence.
10. Write **one** question you have about this module so far — something you want clarified before the summative.

## How this is used

This quiz is **formative** — it's a check-in to see what's landing, not a grade. Teacher reviews answers to adjust pacing for Days 6–10. Students may swap and self-mark.

# Summative Quiz — Subtraction — Nikhilam (All from 9) (Primary)

**When:** end of module (after Day 10). **Duration:** 30 minutes. **Total points:** 20.

## Questions

**Part A — Vocabulary (~5 points)** A1. Define *Nikhilaṁ Navataścaramaṁ Daśataḥ* in your own words. A2. Define *pratiyogi (modern coinage)* in your own words.

**Part B — Concepts (~10 points)** B1. Explain the central idea of Subtraction — Nikhilam (All from 9) in your own words (3–4 sentences). B2. Describe the method or framework students used in Day 4 of this module. B3. Identify ONE source cited in this module and explain why it matters. B4. Compare Subtraction — Nikhilam (All from 9) with a concept from another module (your choice). What's similar, what's different? B5. Identify ONE common misconception about this module and rebut it in 2–3 sentences.

**Part C — Application (~5 points)** C1. Give an example of Subtraction — Nikhilam (All from 9) from your own daily life. C2. Describe one way this module connects to a current real-world issue (or scientific question).

**Part D — Reflection (~5 points; ungraded for content)** D1. What surprised you most in this module? D2. What's one question you still have?

### Marking guidance

- Vocabulary: give credit for correct meaning, even if exact wording differs.
- Concepts: look for evidence of understanding, not memorisation.
- Application: any reasonable example accepted.
- Reflection: ungraded for content; awarded for thoughtfulness.

### Pass mark

A score of 12/20 indicates the student has met the module's learning objectives.

# Homework

# Homework — Subtraction — Nikhilam (All from 9) (Primary)

Light daily tasks. Each takes 5–10 minutes.

1. **Day 1 — The Sūtra — All from 9, Last from 10** · Draw about today's concept. (~5 min)
2. **Day 2 — Complement to 1000 and Beyond** · Draw about today's concept. (~5 min)
3. **Day 3 — Subtract from a Power of 10** · Draw about today's concept. (~5 min)
4. **Day 4 — General Subtraction via Complements** · Draw about today's concept. (~5 min)
5. **Day 5 — Why It Works — The Algebra** · Draw about today's concept. (~5 min)
6. **Day 6 — Bar Model Proof** · Draw about today's concept. (~5 min)
7. **Day 7 — Two's Complement in Computers** · Draw about today's concept. (~5 min)
8. **Day 8 — Speed Day** · Draw about today's concept. (~5 min)
9. **Day 9 — Mixed Practice** · Draw about today's concept. (~5 min)
10. **Day 10 — Final Assessment** · Draw about today's concept. (~5 min)

### Notes

- Homework is intended to consolidate what was learned in class, not introduce new content.
- If a child struggles, the parent should consult the teacher rather than push.
- Capstone project days (Module 15) may have lighter daily homework so students can focus on the project.

# Activity 1 — Complement Flashcards

**Module:** Subtraction — Nikhilam (All from 9) · **Band:** Primary · **Duration:** 20 min

## Purpose

Reinforce the day-by-day concepts of Subtraction — Nikhilam (All from 9) through a hands-on activity. Aligned with NEP 2020 §4.23 (experiential learning).

## Materials

40 flashcards (target − number) pre-made

## What students do

In pairs, one student holds up “1000 − 738” (etc.), the other shouts the answer within 3 seconds using the Nikhilam method.

## Step-by-step

1. **Setup (5 min)** — gather materials; explain the task.
2. **Working phase (10 min)** — students work individually or in pairs.
3. **Share (5 min)** — selected students share their results with the class.

## Differentiation

- **Extension:** ask advanced students to do an additional iteration or analysis.
- **Support:** pair students who need scaffolds; offer partial templates.

## Assessment

This activity is **formative**. The teacher observes participation, accuracy, and reflection quality. No grade is assigned.

## Safety

- General safety briefing before any materials are handled.
- If herbs, plants, or food are involved: kitchen-amount only, no ingestion of unknown preparations.
- Outdoor activities: stay in the designated area; supervised by the teacher.

# Project Brief

# Project Brief — Computer Subtraction Explainer

**Module:** Subtraction — Nikhilam (All from 9) · **Band:** Primary

## What you will do

Make a one-page explainer poster: “How does my calculator subtract?” Show two’s complement subtraction in binary using the same principle as Nikhilam.

## Why

This project asks you to apply concepts from this module to your own life or a real situation. It is graded on the rubric in `rubric.md`.

## Deliverables

- **One-page project plan** (Day 2 of the module): problem, sources, deliverable, timeline.
- **Daily journal entries** (one per day, starting Day 6): what you did, what worked, what didn't.
- **Final artefact** (a drawing / made object).
- **Written reflection** (oral share): what you learned, what surprised you, what you're still uncertain about.
- **Presentation** (3-minute share with the class).

## Timeline

- **Day 1–2:** scope and plan.
- **Day 3–4:** research.
- **Day 5:** mid-project critique.
- **Day 6–7:** build / make.
- **Day 8:** document.
- **Day 9:** rehearse.
- **Day 10:** share day.

## Sources

You must cite at least 1 sources. Use the form: *Author, Title, year.* (See `../sources.md` for examples.)

## Safety

- Any project involving plants / food / outdoor work must be supervised.
- **No herbal medicine recommendations.** Modules 10 and 4 give awareness; therapeutic decisions belong to qualified practitioners.

## Grading

See `rubric.md`. The project is worth the same weight as the summative quiz.

# Rubric

# Assessment Rubric — Subtraction — Nikhilam (All from 9) (Primary)

This rubric applies to the project (see `project-brief.md`) and may also inform the summative quiz.

## Criteria

- **Knows the new word(s)** — 🌟 Strong understanding / 🌱 Growing / 🌧️ Needs more practice
- **Joined in activities** — 🌟 Strong understanding / 🌱 Growing / 🌧️ Needs more practice
- **Asked good questions** — 🌟 Strong understanding / 🌱 Growing / 🌧️ Needs more practice
- **Drew / made something** — 🌟 Strong understanding / 🌱 Growing / 🌧️ Needs more practice

## How the teacher uses this rubric

1. After the project share, score each student on each criterion.
2. Convert to a band (4 = Exceeding, 3 = Meeting, etc.) for the report card.
3. Share at least one strength and one area for growth with each student.

## Self-assessment (Senior students)

Senior students complete a parallel self-assessment using the same rubric before the teacher's assessment. The teacher and student then meet to compare.